

Applied AI & Robotics Portfolio

Academic Journey & Technical Capstones

Cuong Dang

Applied AI and Robotics | Houston City College

<https://github.com/binxixi23/Cuong-Dang-AI-Portfolio>

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Academic Growth & Core Toolkit

- **Journey:** Foundational ML → Advanced Computer Vision → Enterprise AI Deployments
- **Environments:** Python, Google Colab, Jupyter, Git, Windows Command Line
- **Frameworks:** NumPy, Pandas, Scikit-Learn, OpenCV, TensorFlow, Keras, PyTorch



Core Course Capstones

- **Machine Learning:** Telco Customer Churn EDA Project
- **Computer Vision:** SentinelClean Intelligent Background Noise Removal Pipeline
- **AI Applications:** Real-Time Financial Fraud Anomaly Processing and Shielding Engine

Machine Learning: Telco Customer Churn EDA Project



Exploratory Data Analysis,
Predictive Modeling,
Insights

Computer Vision: SentinelCleaner Intelligent Background Noise Removal Pipeline



AI Applications: Real-Time Financial Fraud Anomaly Processing and Shielding Engine



Real-Time Transaction
Analysis, Anomaly Detection,
Immediate Shielding



Developed AI Capabilities

- **Data Pipelines:** Profiling distributions, cleaning messy spaces, feature engineering
- **Vision & Modeling:** Convolutional Neural Networks, YOLO object detection, matrix pixel transformations
- **Analytical Evaluation:** Confusion matrices, IoU scoring, precision-recall optimization, ethical bias mitigation

Horizons & Professional Contact

- **Reflection:** Mastered end-to-end data pipelines, complex model troubleshooting, and deployment strategies
- **Future Goals:** Scaling deep learning frameworks, autonomous robotics integration, edge AI deployments
- **Links:**

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